Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

# SDC Hackathon

# Technical Tools Declaration

**Hosted by Xiamen University Malaysia (XMUM) AI Club**
**Submission Deadline: 25 October 2026**

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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**SDC HACKATHON 2026**

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## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | Milk Dragon |
| **Project / Solution Name** | Orbyte Project Commitment Copilot |
| **Name (Leader)** | Lui Qiu Yang |
| **Email & Phone (Leader)** | luiqiuyang1230@gmail.com ; +6011-1087 1596 |
| **University Affiliation(s)** | Sunway University |
| **Live Prototype URL** | https://hackathon-sdc.replit.app/ |
| **Code Repository URL (public / access granted to judges)** | https://github.com/Qyang04/hackathon-sdc |

## 2. Architecture & Technology Stack

### 2.1 System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

Staff paste quotation text or upload a text-based PDF into a three-step Next.js form. unpdf extracts PDF text, and NVIDIA Nemotron extracts payment milestones for human confirmation. Next.js API routes use Prisma to retrieve synthetic projects, payables, receivables and payroll from PostgreSQL through simulated Orbyte adapters. A deterministic engine calculates daily cash over at least 13 weeks and compares planned, downside and deferred-start scenarios. Fixed rules evaluate cash buffers, thresholds and incomplete data. Nemotron explains computed results without changing figures, rule outcomes or statuses. Failed AI calls allow manual entry or results without an AI summary. Server-side checks restrict decisions to the selected owner role and require the latest assessment and current settings. Demo roles are simulated rather than production authentication. Decisions, assessment snapshots, simulated email notifications and e-sign outcomes are logged. Replit hosts the application, and GitHub hosts its source code. All demonstration business data is synthetic; Orbyte, email and e-sign connections are simulated.

### 2.2 Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| TypeScript – 5.9.3 | Main programming language for frontend components, backend APIs, business logic and scripts. | Apache-2.0 |
| Next.js – 15.5.28 | Full-stack framework providing page routing, server rendering and backend API routes. | MIT |
| React – 19.1.10 | Builds interactive interfaces, forms and reusable UI components. | MIT |
| React DOM – 19.1.10 | Renders React components in the browser. | MIT |

| Name & Version | Purpose in Project | License |
|---|---|---|
| Prisma Client (@prisma/client) – 6.19.3 | Provides typed database access for commitments, assessments, settings and activity logs. | Apache-2.0 |
| Prisma CLI (prisma) - 6.19.3 | Manages the database schema and generates Prisma Client. | Apache-2.0 |
| Zod – 4.1.13 | Validates request inputs, settings and structured AI responses. | MIT |
| unpdf – 1.8.1 | Provides PDF processing and text extraction for quotation documents. | MIT |
| Vitest – 4.1.12 | Runs automated tests for calculations, validation, policies and API behaviour. | MIT |
| tsx – 4.20.6 | Executes TypeScript scripts for database seeding, backups and restoration. | MIT |
| @types/node - 22.20.5 | Supplies Node.js type definitions for TypeScript development. | MIT |
| @types/react – 19.1.17 | Supplies React type definitions for TypeScript development. | MIT |
| @types/react-dom – 19.1.11 | Supplies React DOM type definitions for TypeScript development. | MIT |
| PostgreSQL 17 | Relational database storing application records and synthetic demo data. Local development uses postgres:17-alpine; the live application uses Replit PostgreSQL. | PostgreSQL License |
| Node.js – 20 | JavaScript runtime used to build and run the Next.js application and development tools. | MIT |
| HTML and CSS (web standards) | Page structure, styling, responsive layouts and themes. | N/A |
| Plus Jakarta Sans (font) | Main application interface font, loaded through ‘next/font/google’ | SIL Open Font License 1.1 |
| Newsreader (font) | Serif italic font used for AI explanation text, loaded through ‘next/font/google’ | SIL Open Font License 1.1 |

## 3. APIs & Cloud Services

### 3.1 External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API & Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| NVIDIA-hosted Nemotron API - *nvidia/nemotron-3.5-lightning-30b-a3b* | Extracts quotation milestones and explains computed assessment results through an OpenAI-compatible chat-completions endpoint. | Free NVIDIA API endpoint |
| Orbyte Sales / Accounting / Projects | Seeded local adapters provide quotations and commitments. No live external Orbyte API is called. | N/A - simulated locally; no external API access |
| Orbyte eSign | Records a simulated acceptance/signature outcome after owner approval. No real signing provider is called. | N/A - simulated locally; no external API access |
| Email / Orbyte Connect | Stores simulated notifications in the activity log and Outbox. No message is sent to an external email service. | N/A - simulated locally; no external API access |
| Google Fonts - Google | Downloads Plus Jakarta Sans and Newsreader at build time through Next.js; the application then serves the font files. | Public font delivery; no paid API key |

### 3.2 Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Replit Autoscale - Replit | Hosts the Next.js app and API. Configured to build with npm run build and start with npm run start. Live URL: https://hackathon-sdc.replit.app/ | Sponsor-provided Replit Core access |

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Replit PostgreSQL - Replit | Persistent application database accessed using Prisma and DATABASE_URL. Stores synthetic demo data, decisions and logs. | Replit Core credits |
| GitHub | Hosts source code and dependency lockfile at https://github.com/Qyang04/hackathon-sdc | Source code repository |
| NVIDIA API Catalog - hosted inference | Processes quotation text and computed evidence using NVIDIA Nemotron through "https://integrate.api.nvidia.com/v1" | Free API endpoint |

## 4. AI Tools, Models & Datasets

### 4.1 AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| nvidia/nemotron-3.5-lightning-30b-a3b | NVIDIA (NVIDIA-hosted API) | Pre-trained model for quotation extraction and explanations. No team training or fine-tuning. | OpenMDW License Agreement v1.1; NVIDIA hosted API terms apply |

### 4.2 Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| Synthetic SME commitments | Team-created; src/lib/demo/data.ts in the project repository | Synthetic; running application and demo | Project repository: Apache-2.0 | N |
| Demo quotations and payment milestones | Team-created demo data and synthetic quotation inputs | Synthetic; extraction and workflow demonstration | Project repository: Apache-2.0 | N |
| Automated test fixtures | Team-created tests and docs/test-cases.md in the repository | Synthetic; evaluation against expected results | Project repository: Apache-2.0 | N |
| Training / fine-tuning datasets | None | No team training or fine-tuning performed | N/A | N/A |
| External / real personal datasets | None | Public demo intended for synthetic inputs only | N/A | N |

### 4.3 AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

| **Tool & Version** | **Used For** | **Extent of Use (minor / moderate / major)** |
|---|---|---|
| Claude Code – Opus 5.5 | Assisted with project development, code implementation and organising the coding workflow. | Major |
| Claude – Opus 5.5 | Developed the initial project structure and master plan, including the proposed architecture and workstream breakdown. | Moderate |
| ChatGPT – 6.1 Sol Light and Claude – Opus 5.5 | Assisted with planning coding tasks, organising the development workflow and implementing project code. | Moderate |

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

| |
|---|
| Automated tests cover hand-calculated cash cases, validation, policy outcomes, API permission checks, stale assessments and unsupported amounts in AI briefs. Structured AI output is validated with Zod; returned source snippets must appear in the quotation, and extracted milestones require human confirmation. AI briefs use computed evidence, and unsupported RM amounts are rejected. Failed AI calls leave deterministic results unchanged and allow manual entry. We retain responsibility for reviewing AI-generated code, content and documentation and for explaining the code we submit. |

## 5. Sponsor-Provided Resources

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

| **Sponsor** | **Resource (API / credits / platform / data)** | **How It Was Used in the Project** |
|---|---|---|
| Replit | Sponsor-provided development and deployment platform | Hosts the live prototype using the project build and run configuration. |
| Replit | PostgreSQL and Secrets | Provides PostgreSQL storage and server-side Secrets for database and AI credentials. |

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

Replit provides the sponsor-supported environment for hosting the Next.js prototype, storing demo data in PostgreSQL and managing deployment secrets. The application is configured for Autoscale and can also run locally. Replit is essential to the submission hosting requirement, but the technical implementation can move to another Node.js host and PostgreSQL service. The deterministic engine, validation and owner-review workflow do not depend on Replit-specific business logic.

## 6. Security, Privacy & Responsible Use

### 6.1 Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

The demo stores synthetic quotations, company commitments, payment schedules, assessment snapshots, decisions and activity logs in PostgreSQL. Submitted quotation text is saved and sent to the configured AI API for milestone extraction; computed cash metrics and policy findings are sent for the AI brief. Real personal or confidential business data must not be entered in the public demo. Demo identities and roles are cookie-based simulations, not authenticated users; server permission checks do not provide production access control. AI outputs are schema-validated and extracted milestones require confirmation. API credentials are configured through environment variables/Replit Secrets rather than client-side code. Records remain until demo reset or manual deletion; no automatic retention expiry is implemented. Database backups require separate deletion. NVIDIA provides hosted AI inference. The prototype is not ready to process real personal data.

### 6.2 Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

AI can misread amounts or dates, omit milestones, hallucinate explanations or follow malicious quotation text. Schema validation, source-snippet checks, human confirmation and deterministic calculations limit the impact. Briefs receive computed evidence, and unsupported RM amounts are rejected; this does not guarantee every statement or date is correct. Extraction has a 30-second timeout and briefs 20 seconds; errors leave manual entry or calculated results available. Provider rate limits, availability and retention depend on the configured service. Text PDFs are supported; scanned documents/OCR are not. Synthetic data and simulated Orbyte, email and e-signing do not prove real integrations. Cookie-based role switching is a demo mechanism, not production authentication. Decisions support human review and are not financial advice.

## 7. Compliance & Readiness Checklist

*Tick all boxes that apply before submitting.*

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [x] The prototype described here is accessible to judges through the links provided in Section 1.
- [x] All fields in this template have been completed within the word limits stated.

## 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

**Yang**

24/10/2026